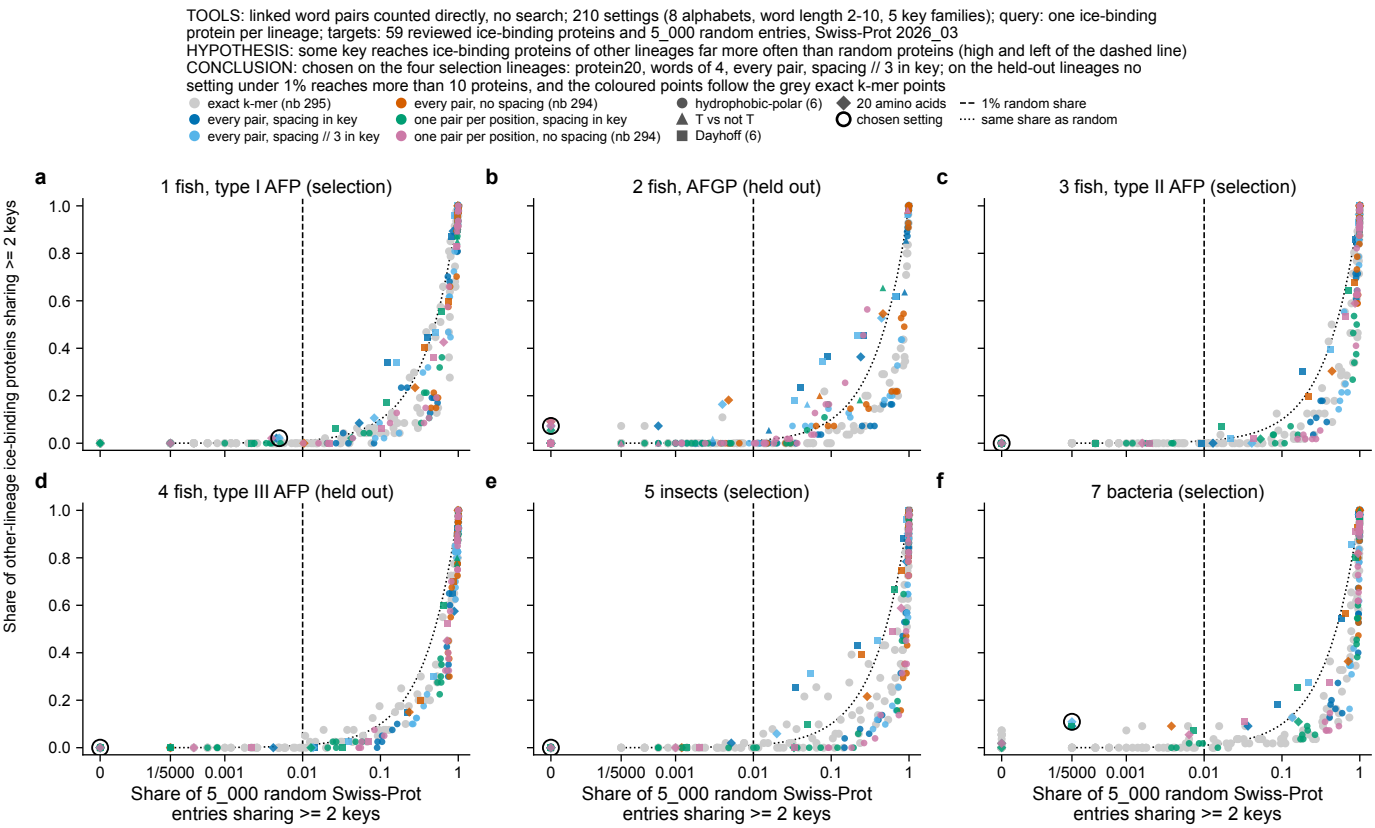

TOOLS: linked word pairs counted directly, no search; 210 settings (8 alphabets, word length 2-10, 5 key families); query: one ice-binding protein per lineage; targets: 59 reviewed ice-binding proteins and 5_000 random entries, Swiss-Prot 2026_03
HYPOTHESIS: some key reaches ice-binding proteins of other lineages far more often than random proteins (high and left of the dashed line)
CONCLUSION: chosen on the four selection lineages: protein20, words of 4, every pair, spacing // 3 in key; on the held-out lineages no setting under 1% reaches more than 10 proteins, and the coloured points follow the grey exact k-mer points
exact k-mer (nb 295)
every pair, spacing in key
every pair, spacing // 3 in key
every pair, no spacing (nb 294)
one pair per position, spacing in key
one pair per position, no spacing (nb 294)
hydrophobic-polar (6)
T vs not T
Dayhoff (6)
20 amino acids
chosen setting
1% random share
same share as random
a
1 fish, type I AFP (selection)
b
2 fish, AFGP (held out)
c
3 fish, type II AFP (selection)
d
4 fish, type III AFP (held out)
e
5 insects (selection)
f
7 bacteria (selection)
Share of other-lineage ice-binding proteins sharing >= 2 keys
0.0
0.2
0.4
0.6
0.8
1.0
0
1/5000
0.001
0.01
0.1
1
Share of 5_000 random Swiss-Prot entries sharing >= 2 keys